

7/2/2014

Mr. Dave Schipper
Department of Environmental Quality
350 Ottawa N.W.
Grand Rapids, MI 49503

Subject: Amendment to Existing RMP

Dear Mr. Schipper:

The City of Cadillac would like to add a method to Section III (Land Application Plan) Part B-4 of our RMP whereas we can surface apply biosolids. We would like to have this option, if the sub-surface injection cannot be carried out. We meet the criteria for “Exceptional Quality” Biosolid as listed in Part 24;

(i) Pollutant ceiling concentrations in R 323.2409(5)(a)

(ii) Pollutant concentrations in R 323.2409(5)(c)

(iii) Vector attraction reduction options in R 323.2415(4) (a); *we had 54.11% volatile reduction in 2013 with primary digester heated to average of 96.20 °F using external heat exchanger and retention time of 22.2 days. And class A pathogen reduction alternatives in R 323.2414(2)(a); the density of salmonella sp. bacteria in the biosolids was <3 MPN per 4 grams of total solids (dry weight basis) for 2013.*

The following analytical data represents the average contents of biosolids in our most recent analysis. Please find the attached documents for the complete list of the variables analyzed.

Recent biosolids analysis in mg/kg (dry weight basis)

Name	Concentration (mg/kg) Dry Weight	Pollutant Concentrations (Table 3, 40 CFR 503.13) (monthly average)	Ceiling Concentrations* (Table 1, 40 CFR 503.13) (daily maximum)
Arsenic	15.3	41 mg/kg	75 mg/kg
Cadmium	2.7	39 mg/kg	85 mg/kg
Copper	706	1500 mg/kg	4300 mg/kg
Lead	<50	300 mg/kg	840 mg/kg
Mercury	0.72	17 mg/kg	57 mg/kg
Molybdenum	23	N/A**	75 mg/kg
Nickel	18.6	420 mg/kg	420 mg/kg
Selenium	4.2	100 mg/kg	100 mg/kg
Zinc	1060	2800 mg/kg	7500 mg/kg
Nitrogen Concentration		N/A	N/A

* Biosolids may not be land applied if any pollutant exceeds these values.

** EPA has temporarily removed molybdenum limits from Table 2, Table 3 and Table 4

If you have any questions, please do not hesitate to contact me.

Sincerely,

Hamid Rahbarnoohi, Water Resources Supervisor

City of Cadillac

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